

ENAE 410 – Design Lab 1

Predicting Flight Performance, Aircraft Sizing

Objective

The objective of this lab is to teach students how to size a model aircraft by matching the propulsion system and airframe for "flyability" and maneuverability, using provided thrust, lift, and drag curves.

Background

1.1 Motivation

A key aspect of aircraft design, regardless of scale or application, lies in achieving an optimal "matching" between the propulsion system and the airframe. Even in the design of small remote-controlled (RC) aircraft and drones, this balance is crucial to ensure successful flight and maneuverability. In the design project for this course, you will be provided with a fixed propulsion system, the E-Flite Park 250, and your system weight, which includes essential components such as the electronic speed control, servos, receiver, battery, and wires, will also be predetermined. The primary objective of this lab is to explore how to effectively match the size and shape of the aircraft, based on a reference model, to the fixed propulsion system. By achieving this optimal "matching," you will ensure that the aircraft exhibits key performance metrics necessary for successful flight and maneuverability.

The significance of this activity lies in its real-world relevance. In the field of aviation, engineers and designers often encounter the challenge of selecting the appropriate propulsion system and airframe configuration for a given aircraft. This process involves considering various performance metrics, weight constraints, and operational requirements. By engaging in this lab exercise, you will gain valuable insights into the intricacies of aircraft design and the importance of selecting the right combination of components for a functional and agile flying machine. Moreover, understanding the principles of matching the propulsion system and airframe will not only be beneficial for the design project in this course but also lays the foundation for your future endeavours in aviation or aerospace engineering. The ability to make informed decisions regarding aircraft sizing and system integration is a critical skill that will serve you well in your academic and professional pursuits.

In conclusion, this lab provides a valuable foundation and hands-on experience in understanding the critical concept of "matching" the propulsion system and airframe for a successful aircraft design. The knowledge and insights gained from this activity will play a pivotal role in your upcoming term project, where you will have the exciting opportunity to design your own RC model airplane. By applying the principles learned in this lab, you will be equipped with the necessary tools to make informed decisions in selecting the appropriate components, optimizing the aircraft's performance metrics, and ensuring the aircraft's "flyability" and maneuverability. The skills developed during this lab will directly contribute to the success of your term project, allowing you to create a well-designed and functional model airplane that meets all performance requirements.

2 Technical information

2.1 Reference geometry aerodynamics

To support the class, a reference geometry has already been put through a wind tunnel study. The reference geometry is shown in the following figure:

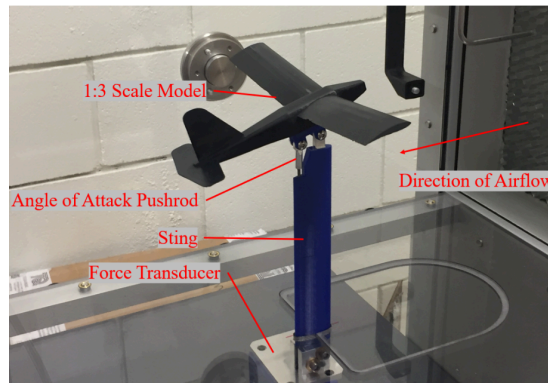


Figure 1: Experimental Setup in Undergraduate Wind Tunnel

The wind tunnel data is shown in the following figures:

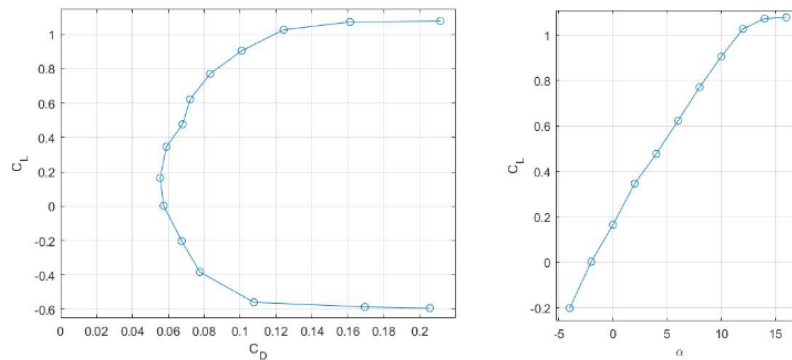


Figure 2: Experimental Lift and Drag Coefficients

Table 1: Lift and Drag Data for Reference Model

α	C_L	C_D
-4.000	-0.206	0.067
-2.000	0.000	0.057
0.000	0.162	0.055
2.000	0.344	0.059
4.000	0.473	0.068
6.000	0.620	0.072
8.000	0.769	0.083
10.000	0.907	0.101
12.000	1.026	0.124
14.000	1.069	0.161
16.000	1.075	0.211

2.2 Thrust and power information

Theoretical thrust curve for the 6x4 propeller on the E-flite Park 250:

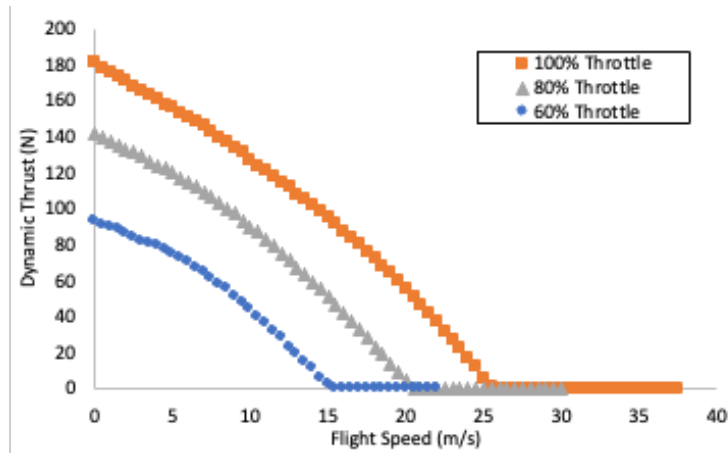


Figure 3: Theoretical Motor/Propellor Performance

An excel sheet is posted on D2L which contains this data. These curves were produced by an affordable commercial software called MotoCalc. This software is extremely useful for hobbyists and designers of RC airplanes and drones.

2.3 Component masses

For the calculations in this lab, please assume these components are required:

- Servos: 5.1 grams (Assume you need 3 of these (1 for ailerons, 1 for rudder, 1 for elevator))
- Motor: (E-flite Park 250) 18.3 grams
- 20A speed control: 16.6 grams
- Receiver: 7.2 grams
- Landing gear: 1.7 grams
- Propeller 5.1 grams
- Battery: 30.7 grams

2.4 Estimating aircraft mass

Consider how the mass of the aircraft will scale with changing size. For a new span, you can compute a new surface area from:

$$SA = SA_{ref} \left(\frac{b^2}{b_{ref}^2} \right)$$

Similarly, the volume can be computed by:

$$Vol = Vol_{ref} \left(\frac{b^3}{b_{ref}^3} \right)$$

The reference areas and volume are based on the wind tunnel model. These are given as:

$$SA_{ref} = 0.02207 \text{ m}^2$$

$$Vol_{ref} = 3.8292 \times 10^{-5} \text{ m}^3$$

$$b_{ref} = 0.19948 \text{ m}$$

Assume that the density of the lightweight PLA which we will use for construction is 800 kg/m^3 . With this information, we can assume that the mass of the airplane can be calculated with:

$$m = \%infill \times \rho_{material} \times Vol + \rho_{material} \times wallthickness \times SA$$

The total mass of the airplane is the:

$$m_{total} = m_{material} + m_{components}$$

2.5 Wing area

The wing area is also scaled with the square of a reference length ratio.

$$S_w = S_{w,ref} \left(\frac{b}{b_{ref}} \right)^2$$

This wing area will be used to compute the lift on the scaled aircraft. The reference wing area is:

$$S_{w,ref} = 0.00631753 \text{ m}^2$$

3 Analysis: Sizing the Model Aircraft

In this analysis section, we will focus on sizing the model aircraft using the content and calculations covered in class to ensure it is "flyable" and performs optimally under different flight conditions. To achieve this, the following key calculations and figures should be presented in your lab report.

1. **Flight Performance Calculations:** Perform flight envelope calculations to determine critical parameters, such as the minimum and maximum speeds required for safe flight and the maximum bank and climb angles the aircraft can sustain. Plot graphs comparing the required thrust (based on the flight envelope) to the available thrust of the aircraft at various flight speeds. These plots will provide valuable insights into the adequacy of the aircraft's propulsion system. Ensure the figures are well-labelled and easy to interpret.
2. **Stall Speed Calculation:** Determine the stall speed of the model aircraft based on the lift curve. This speed represents the minimum speed required to maintain lift and avoid stalling during flight. Hint: This is computed at Cl_{max} .

3. **Graphical Representations:** Create clear and informative graphs to visualize the relationships between different parameters, such as speed, thrust, and stall speed. These graphs should support your analysis and demonstrate the aircraft's performance characteristics.
4. **Discussion of Results:** After presenting the calculations and figures, discuss the implications of your findings. Address any limitations or assumptions made during the analysis and how they might impact the aircraft's performance.
5. **Design Recommendations:** Based on your analysis, make design recommendations for the model aircraft to ensure it meets the "flyable" criteria and achieves optimal performance. Discuss any necessary adjustments or improvements to enhance its flight capabilities.

By focusing on flight envelope calculations and plotting graphs comparing required to available thrust, you will demonstrate a comprehensive understanding of sizing the model aircraft and its propulsion system's suitability. Make sure to organize your analysis logically and use appropriate labels, units, and references for a clear and professional lab report. A template lab report is also available on D2L.

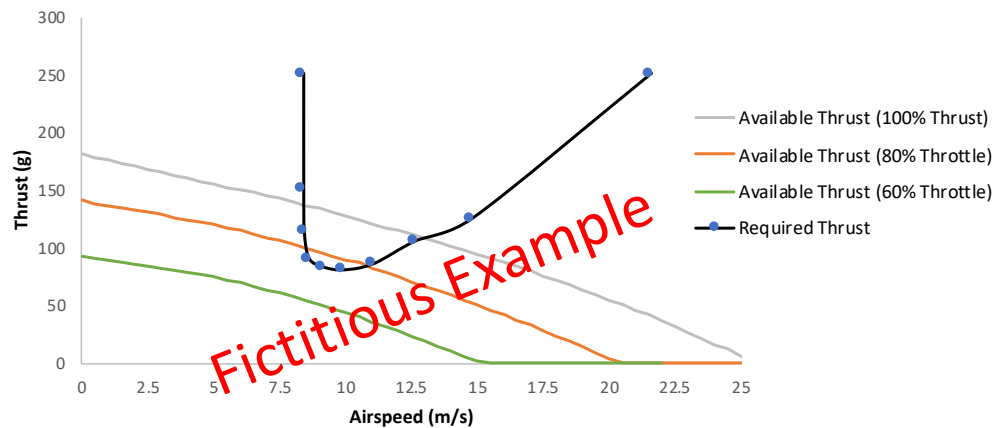


Figure 4: Example Graphical View of Aircraft Flight Speed

4 Summary and Expected Outcomes

By completing this lab, you should learn the basics of how available thrust, component mass, and aerodynamics combine in order to determine an aircraft's performance. The goal is to extend this analysis to your own design. In the subsequent labs, you will learn special tools which will allow you to predict aerodynamics yourself, which you will then use to complete your design.

5 References and supporting materials

The following may be consulted for additional information:

1. Course notes, slide deck, videos (available on D2L)
2. Aerodynamics and Aircraft Performance, Marchman, PDF available: <https://vtechworks.lib.vt.edu/handle/10919/96525>

Appendix A

An example of a commercial plane with the Park 250



Wingspan:	30 in (760mm)
Overall Length:	23 in (595mm)
Wing Area:	310 sq in (20.0 sq dm)
Flying Weight:	7.5–8 oz (210–225 g)
Motor Size:	Park 250
Radio:	3 channels with 2 Super Sub-Micro servos
Servos:	(2) S60's
Prop Size:	GWS 6x5 slow flyer
Hardware Included:	yes
Speed Control :	9- to 10-amp brushless
Recommended Battery:	7.4 V 730mAh 2-cell Li-Po
Approx. Flying Duration:	15 minutes
Scale:	Indoor

<https://www.skycrafthobbies.com/eflite/438>

E-Flite Park 250 Motor Specifications

Park 250 Brushless Outrunner Instructions

Thank you for purchasing the E-flite Park 250 Brushless Outrunner motor. This motor is ideal for 3D models 4-6 oz (115-170g), and sport and scale models 6-12 oz (170-340g) or models requiring up to 55 watts of power.

Park 250 Brushless Outrunner Features:

- Ideal for 3D models 4-6 oz (115-170g) and sport and scale models 6-12 oz (170-340g)
- Ideal for up to 55 watts
- Integrated prop saver design
- Designed for firewall mount or carbon tube mount
- Extremely quiet and lightweight - just .5 ounces (14 grams)
- Includes mount, mounting hardware, prop saver adapters, prop adapter, and 2mm gold BL connectors
- Great lightweight outrunner for small indoor foamies
- Upgraded performance motor for indoor IPS and 180 motors
- High quality construction with ball bearings and hardened steel shaft
- Slotted 12-pole outrunner design

Examples of Airplane Setups:

Indoor Profile 3D Airplane

Weight w/Battery: 4.3 oz (122g)
Prop: 6 x 5 Slow Flyer
Battery: 480mAh 2-Cell 7.4V Li-Po (THP4802SJPL2)
ESC: 10A Brushless

Comments – Extreme 3D performance. Will hang on the prop at ½ stick and when you add full throttle it takes off straight up like a rocket.

Jenny JN-4 Slow Flyer 250 (EFL1900)

Weight w/Battery: 8.2 oz (232g)

Prop: 7 x 4 Slow Flyer
Battery: 480mAh 2-Cell 7.4V Li-Po (THP4802SJPL2)
ESC: 10A Brushless

Comments – Great power for scale slow flying performance.